ModelIt!

Teacher's Guide

Why Do Animals Migrate Thousands of Miles?

Modeling the Costs and Benefits of Long-Distance Animal Migration

Quick Reference

Grade Level	Grade 10L2
NGSS Standard	HS-LS2-8, HS-LS4-6
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

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1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-LS2-8, HS-LS4-6

Evaluate evidence for the role of group behavior on individual and species' chances of survival and reproduction. Create or revise a simulation to test a solution to mitigate adverse impacts of human activity on biodiversity.

Three-Dimensional Learning

Science Practice	**Constructing Explanations and Designing Solutions** — Students construct evidence-based explanations of why migration evolved and design conservation solutions to protect migration corridors from climate change and habitat loss.
Disciplinary Core Idea	**LS2.D Social Interactions and Group Behavior / LS4.C Adaptation** — Group behaviors such as collective migration enhance individual survival and reproductive success; natural selection has shaped migration as an adaptive response to seasonal resource variation.
Crosscutting Concept	**Cause and Effect** — Students trace the causal chains from environmental triggers through physiological responses and behavioral decisions to population-level outcomes, identifying how disruption at any link can collapse the entire migration system.

CAST Testing Alignment

MULTIPLE CHOICE:

Climate change causes spring insect emergence to occur two weeks earlier at a bird's breeding ground, but the bird's migration departure is triggered by day length, which has not changed. Based on your model, what is the most likely consequence?

CONSTRUCTED RESPONSE:

Using your model, explain why the destruction of a single stopover wetland in Delaware could contribute to the decline of a shorebird species that breeds in the Arctic and winters in South America. Reference Energy Reserves, Food Availability, and Reproductive Success in your explanation.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Every year, billions of animals undertake journeys that would be extraordinary even with modern technology. Arctic terns fly from pole to pole — 71,000 kilometers annually. Bar-tailed godwits fly 11,000 km non-stop across the Pacific Ocean without resting. Monarch butterflies navigate 4,000 km to a specific mountain forest in Mexico that no individual butterfly has ever visited before. Migration is one of evolution's most remarkable achievements — and one of its most fragile.

Why Migrate? The Cost-Benefit Equation

Migration is extraordinarily dangerous — an estimated 50-80% of some migratory bird species die during migration each year. So why do it? Because the reproductive payoff exceeds the survival cost. High-latitude breeding grounds offer longer daylight hours for foraging, explosive seasonal food production (insect hatches, plant blooming), fewer parasites and diseases, and less competition. A bird that migrates to the Arctic and raises a clutch of 6 chicks — even if 50% die on the return migration — still produces more surviving offspring than a sedentary relative in the tropics that raises 2 chicks. Natural selection favors the strategy with the highest net reproductive output, even if individual risk is extreme.

Navigation: A Biological GPS

Migratory animals use a multi-modal navigation system of astonishing sophistication. The magnetic compass uses magnetite crystals or cryptochrome proteins to detect Earth's magnetic field direction and intensity. The celestial compass uses the sun's position (time-compensated by an internal circadian clock) during the day and star patterns at night. The olfactory map uses chemical gradients in air or water (salmon use this to find their natal stream). Landmarks and memory allow experienced migrants to recognize landscape features. The 'many wrongs' principle explains collective navigation: individual birds make small navigation errors in random directions, but averaging across a flock cancels errors and produces a more accurate group trajectory than any individual could achieve alone.

The Phenological Mismatch Crisis

Migration timing evolved to synchronize with seasonal food availability. Arctic-breeding birds arrive when insect populations explode. Whales arrive when krill blooms occur. This synchronization developed over millions of years of coevolution. Climate change is disrupting this synchronization because different components of the system respond to different cues. Plants and insects respond to temperature — which is shifting earlier. But many migratory animals respond to day length (photoperiod) — which does not change with climate. The result is phenological mismatch: birds arrive on their genetically programmed schedule only to find that the food peak occurred two weeks ago. Populations that depend on this timing are declining rapidly.

Stopover Ecology: The Critical Link

A migratory shorebird flying from Argentina to the Arctic cannot carry enough fat to make the trip in one flight. It must stop at specific locations to rest and refuel — often for just 2-3 weeks, doubling its body weight before continuing. Delaware Bay, for example, hosts 1 million shorebirds each May that depend on horseshoe crab eggs to fuel the final leg to the Arctic. If Delaware Bay is degraded, the entire flyway population collapses — even though breeding and wintering habitats are intact. The chain is only as strong as its weakest link, and stopover sites are increasingly threatened by coastal development, pollution, and climate change.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify seven components of the migration system: Navigation Accuracy, Energy Reserves, Climate Trigger, Predator Avoidance, Group Size, Food Availability, and Reproductive Success — with Climate Trigger and Food Availability as external environmental conditions.

E **Establish Relationships**

Students determine that Climate Trigger initiates migration timing, but Food Availability at the destination determines whether arriving migrants can successfully breed. Energy Reserves must be sufficient for the journey, Navigation Accuracy determines route efficiency, and Group Size affects both navigation and predator survival. Reproductive Success is the ultimate outcome.

V **Visualize & Model**

Students build a computational model showing how the timing and spatial requirements of migration must align for populations to sustain themselves across generations.

E **Evaluate Outcomes**

Students run optimal migration, climate mismatch, and habitat fragmentation scenarios to determine which disruptions are most damaging to population viability.

R **Revise & Extend**

Students add Light Pollution, Wind Pattern Shifts, or Genetic Program Fidelity to explore how additional human-caused and evolutionary factors affect migration success.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"A bird the size of your fist just flew 11,000 kilometers non-stop across the Pacific Ocean. No food. No rest. No GPS. How?"*

Watch For

Show the bar-tailed godwit non-stop Pacific crossing data on a map. Let the distance sink in. Ask: Could any human-built drone do this?

Slide 2: Learning Objectives

3 min

Say: *"Today you're modeling one of nature's most extreme behaviors — and asking why natural selection would favor such a dangerous strategy."*

Watch For

Have students read objectives. Pre-teach 'magnetoreception' and 'phenological mismatch.' Quick-write: Why would an animal choose the most dangerous possible way to live?

Slide 3: Big Question

3 min

Say: *"Migration kills 50-80% of some species every year. By any common-sense measure, it seems insane. But evolution doesn't do insane — it does math. What's the equation?"*

Watch For

Students calculate: If migration kills 60% of birds but survivors produce 6 chicks, while staying put kills 10% but produces only 2 chicks — which strategy wins? (Migration: 0.4 x 6 = 2.4 offspring vs. Sedentary: 0.9 x 2 = 1.8 offspring.)

Slide 4: LEVER Framework

2 min

Say: *"We're modeling the most dangerous commute in the natural world — and figuring out how climate change is about to break it."*

Watch For

Preview each LEVER step. Emphasize that migration is a cost-benefit optimization refined by millions of years of evolution.

Slide 5: Activity 1: Components

8 min

Say: *"Seven factors determine whether a migration succeeds or fails. Two come from the environment. The rest come from the animal and its flock."*

Watch For

Guide component sorting. Discuss why Climate Trigger and Food Availability are external (environmental conditions) while biological capabilities are internal responses shaped by evolution.

Slide 6: Activity 2: Connections

8 min

Say: *"The migration system is a precision machine with zero margin for error. Change one thing — timing, fuel, route, food — and the whole system can collapse."*

Watch For

Help students map the connections. Identify why Energy Reserves at stopover sites are the bottleneck and how Group Size improves multiple components simultaneously.

Slide 7: Activity 3: Simulation

10 min

Say: *"Three scenarios: perfect migration, climate mismatch, and lost stopovers. Your model will show which one is most devastating."*

Watch For

Students predict outcomes. Run all three scenarios. The phenological mismatch scenario often surprises students — it's more damaging than they expect because it undermines Reproductive Success, not just survival.

Slide 8: Discoveries

5 min

Say: *"Migration is not about strength or speed — it's about timing, fuel, and precision. And we're disrupting all three simultaneously."*

Watch For

Lead discussion connecting model findings to real species declines. Show satellite tracking data of actual migratory routes. Discuss which species are most vulnerable.

Slide 9: STEM Challenge

12 min

Say: *"A shorebird population has crashed 70%. The stopovers are gone. Design the conservation plan that saves them — across international borders."*

Watch For

Groups design corridor conservation plans specifying which habitats to protect, how to coordinate across countries, and how to monitor success. Present as proposals to a simulated conservation board.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Climate Trigger, Food Availability
- **INTERNAL:** Navigation Accuracy, Energy Reserves, Predator Avoidance, Group Size, Reproductive Success

Relationships

- **Climate Trigger to Energy Reserves:** POSITIVE (+) — Appropriate Climate Trigger timing initiates migratory preparation, including hyperphagia (excessive eating) that builds fat reserves for the journey. Birds can double their body weight in 2-3 weeks of pre-migration feeding, converting food to the high-energy fat stores needed for sustained flight.
- **Energy Reserves to Navigation Accuracy:** POSITIVE (+) — Adequate Energy Reserves allow migrants to fly at optimal altitudes and speeds for navigation rather than being forced into energy-saving behaviors that compromise route accuracy. Well-fueled birds can maintain precise compass headings; exhausted birds make more navigation errors and may be forced to land in suboptimal locations.
- **Food Availability to Reproductive Success:** POSITIVE (+) — Higher Food Availability at breeding grounds directly increases Reproductive Success by providing the caloric resources needed for egg production, nestling growth, and adult survival during the breeding season. When food peaks align with arrival timing, clutch sizes are larger and chick survival rates are higher.

Key Discoveries

- Migration persists because the Reproductive Success at distant destinations exceeds the mortality cost of the journey — it's an evolutionary cost-benefit calculation that has been refined over millions of years
- Group Size provides compound benefits: collective navigation averaging (the 'many wrongs' principle), predator dilution, aerodynamic drafting, and social information about food and danger
- Phenological mismatch from climate change is the most insidious threat to migration because it disrupts the timing coordination that migration depends on — animals arrive on schedule but food does not
- Stopover sites are the 'gas stations' of migration — losing even a single critical stopover can cause population collapse across the entire route, making habitat at rest stops as important as habitat at destinations

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Animals migrate to escape cold weather"	While temperature plays a role, migration is primarily driven by food availability, not cold avoidance. Many migratory birds can physiologically tolerate cold temperatures — they leave because food resources (insects, nectar, small mammals) disappear in winter, not because it's too cold. This is why some species migrate to tropical regions that are actually warmer than necessary for survival — they're following the food, not the thermostat.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Migration routes are learned from parents"	While some species (geese, cranes) learn routes by following experienced adults, many migrants are entirely self-navigating on their first trip. Juvenile cuckoos, raised by foster parents of a different species, migrate to Africa alone — having never seen the route and without any adult cuckoo to follow. Their navigation is genetically encoded: the direction, distance, and timing are inherited, not learned.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Human activity doesn't affect migration much"	Human activity is the primary threat to migratory species worldwide. Light pollution kills an estimated 1 billion birds annually in the US alone by disorienting nocturnal migrants. Wind turbines and communication towers kill hundreds of millions more. Coastal development destroys critical stopover wetlands. Climate change disrupts timing synchronization. Agricultural conversion eliminates grassland and forest habitat along routes. Migratory species are declining 2-3 times faster than sedentary species precisely because they depend on a chain of habitats across thousands of miles — and humans are degrading links across the entire chain.